

TANVISHA KOSE

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SUMMARY

Data Scientist with nearly 3+ years of experience building Python, SQL, and statistical modeling solutions across structured and text data. Develops predictive, classification, forecasting, anomaly detection, NLP, and generative AI workflows, with expertise in feature engineering, model evaluation, data validation, and cloud deployment through AWS SageMaker and Azure ML. Communicates quantitative findings clearly through analytical reporting and decision-support solutions.

TECHNICAL SKILLS

Programming & Data Processing: Python, SQL, SAS PROC SQL, Pandas, NumPy, Data Cleaning, Data Wrangling, Exploratory Data Analysis, Feature Engineering, Data Validation, Data Reconciliation, R

Machine Learning & Statistical Analytics: Predictive Modeling, Classification, Regression, Clustering, Time-Series Forecasting, Anomaly Detection, Statistical Analysis, Model Evaluation, XGBoost, Random Forest, SHAP, Walk-Forward Validation, Biostatistics

Generative AI & NLP: NLP, Generative AI, LangChain, Retrieval-Augmented Generation, Agentic RAG, Embeddings, Vector Databases, Semantic Search, Ranking and Reranking, Prompt Engineering, Whisper, Machine Translation, Text-to-Speech

Data Engineering & Platforms: Databricks, PySpark, Apache Spark, Airflow, ETL Pipelines, Data Integration, Data Transformation, Batch Processing, Distributed Data Processing, REST APIs, Data Quality Monitoring

Cloud, MLOps & Deployment: AWS (SageMaker, S3, EC2, CloudWatch), Microsoft Azure (Azure Machine Learning, Blob Storage, Azure Container Registry, Application Insights), Docker, CI/CD Pipelines, Model Deployment, Model Drift Monitoring, Workflow Automation

Business Intelligence & Applied Analytics: Power BI, Excel, Dashboard Development, KPI Reporting, Demand Forecasting, Pricing Analytics, Predictive Maintenance, Procurement Analytics, Supplier Performance Analysis, Inventory Analysis, Root Cause Analysis, Master Data Management

PROFESSIONAL EXPERIENCE

Course5 Intelligence

Jan 2026 - Present

Data Scientist

New Brunswick, NJ, USA

- Develop Python and SQL data pipelines in Databricks using PySpark, Pandas, and Airflow, improving processing efficiency by 28% while integrating customer, transaction, and text datasets.
- Build NLP and Generative AI workflows with LangChain, embeddings, vector retrieval, and prompt engineering, increasing grounded response relevance by 18% across enterprise knowledge assistants.
- Train classification, regression, clustering, and time-series models in Python and R, applying statistical analysis to segmentation, churn, demand forecasting, and pricing analytics and improving predictive accuracy by 14% across client decision-support use cases.
- Design agentic RAG pipelines combining semantic and lexical search, query processing, ranking, reranking, grounding, and evaluation to improve relevance, traceability, and answer trustworthiness.
- Deploy containerized machine-learning services through Azure ML, Docker, REST APIs, and CI/CD, monitor model drift, and present operational insights through Power BI dashboards.

Sonaflex Industries

Jun 2023 - Jun 2024

Data Scientist

Vadodara, GJ, India

- Built anomaly detection and predictive maintenance models using Python and AWS SageMaker, improving monitoring accuracy by 35% across IoT-driven manufacturing operations and equipment datasets.
- Automated SQL and Python-based feature engineering and data transformation workflows, improving processing efficiency by 40% across ERP and manufacturing analytics pipeline operations.
- Developed statistical validation and cost forecasting models using SAS PROC SQL, supporting variance analysis and operational reporting across manufacturing business datasets.
- Generated stakeholder-facing analytical reports summarizing forecasting insights, production trends, and manufacturing performance metrics, reducing reporting turnaround time by 30% and supporting business decision-making initiatives.
- Collaborated with engineering and analytics teams to validate production datasets and support scalable machine learning deployment workflows within manufacturing analytics environments.

Procmart

Apr 2022 - May 2023

Data Analyst

Noida, UP, India

- Analyzed procurement, supplier, product, and order data using SQL and Excel, improving reporting accuracy by 14% across MRO sourcing and fulfillment operations workflows.
- Built Power BI dashboards tracking spend, purchase orders, supplier performance, inventory movement, and delivery KPIs, reducing manual reporting effort by 18% for business teams.
- Performed data validation, reconciliation, and root-cause analysis across ERP and procurement datasets, cutting duplicate and incomplete records by 12% within one year of operations.
- Partnered with procurement, supply chain, sales, and technology teams to translate business requirements into reports, data models, and actionable process recommendations for stakeholders.
- Supported product master data, pricing, supplier catalog, and order lifecycle analysis, ensuring consistent item hierarchies, catalogue accuracy, and reliable decision-making across e-B2B operations.

PROJECTS

- Agentic RAG Knowledge Assistant**

Nov 2025 - Mar 2026

 - Engineered a Generative AI Agentic RAG architecture combining vector-based retrieval, tool calling, and autonomous task planning to answer complex user queries across large-scale enterprise knowledge repositories.
 - Built specialized NLP agents for document retrieval, query decomposition, reasoning, and response validation, enabling the system to dynamically adapt retrieval strategies and generate contextually relevant responses from thousands of indexed documents.
 - Enhanced model reliability through re-ranking, citation-aware generation, and answer-verification workflows that helped control unsupported responses while improving retrieval precision, response quality, and user trust.
- Speech-to-Speech Translation - Cascading Connector Architecture**

Aug 2025 - Nov 2025

 - Architected a multimodal machine-learning pipeline integrating Whisper ASR, machine translation, and text-to-speech models through a cascading connector architecture to support multilingual speech translation workflows.
 - Engineered cross-modal embedding alignment connectors and benchmarked multilingual translation models, improving training efficiency, inference scalability, and BLEU translation performance significantly.
- Tesla Stock Price Prediction and Sentiment Analysis**

Mar 2025 - May 2025

 - Engineered an NLP and machine learning forecasting pipeline integrating technical indicators and sentiment signals from Reddit, Guardian, and NYT APIs successfully.
 - Trained XGBoost and Random Forest forecasting models using walk-forward validation and SHAP analysis, improving interpretability and stock trend prediction performance significantly.

EDUCATION

- Rutgers University**

Sep 2024 - May 2026

Master of Science, Data Science

NJ, USA

 - GPA: 3.5
- Ram Rao Adik Institute of Technology**

Aug 2019 - May 2023

Bachelor of Engineering, Electronics

Mumbai, India

 - GPA: 3.87

CERTIFICATION

- AWS Certified Cloud Practitioner: AWS
- Software Engineering Job Simulation: JPMorgan Chase & Co.
- CompTIA security: Udemy